



AKDENİZ UNIVERSITY
Faculty of Engineering · Department of Computer Engineering
CSE 413 — Optimization Theory and Applications
Midterm Examination · 2026–2027 Fall Semester



EXAMPLE PAPER. Written for self-study. It was not issued by the instructor and is not a real past exam.

Name Surname		Student No.	
Date	___ / ___ / 20___	Signature	
Duration	90 minutes	Total	100 points

Instructions

1. Closed book, closed notes. A non-programmable calculator is allowed.
2. Show all work. A correct answer without justification receives no credit.
3. Answer in the space provided. If you need more space, use the back of the page and say so.
4. Write fractions exactly where possible (e.g. $1/6$, not 0.1667).
5. Mobile phones must be switched off and put away.

Q	1	2	3	4	5	6	7	Total
Max	10	20	15	15	15	10	15	100
Score								

Question 1 — Modeling

10 points

A small furniture workshop makes **desks** and **bookshelves**. Every product goes through a CNC machine and then through manual assembly. The weekly data are:

	Desk	Bookshelf	Weekly capacity
CNC time (h/unit)	2	1	100 h
Assembly time (h/unit)	1	1	80 h
Profit ($\times 100$ TL/unit)	40	30	—

At most 40 desks can be sold per week. Bookshelves sell without limit. The workshop wants to maximize weekly profit.

- (a) Define the decision variables and write the complete mathematical model (objective and all constraints). *6 pts*

- (b) Classify the model: linear or nonlinear? Continuous or discrete? Is the feasible set convex? Justify each answer in one sentence. *4 pts*

Question 2 — The Simplex Method

20 points

Use the model from Question 1:

$$\text{maximize } z = 40x_1 + 30x_2$$

$$\text{subject to } 2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

(a) Convert the problem to standard (equality) form using slack variables s_1, s_2, s_3 . Write the initial basic feasible solution.

4 pts

(b) Solve the problem with the tableau simplex method. Use the *most negative reduced cost* rule to choose the entering variable and the minimum-ratio test to choose the leaving variable. Show every tableau and mark the pivot element each time.

12 pts

(c) State the optimal solution (x_1^*, x_2^*) and the optimal profit. Which constraint is *not* binding at the optimum?

Confirm your answer graphically by listing the vertices of the feasible region and their objective values.

4 pts

Question 3 — Duality and Sensitivity

15 points

Consider again the LP of Question 2.

- (a) Write the dual LP. Use y_1, y_2, y_3 for the dual variables of the CNC, assembly and demand constraints. 5 pts

- (b) Using **complementary slackness** and your primal optimum from Q2(c), find the dual optimal solution y^* .
Verify strong duality. 6 pts

- (c) A subcontractor offers extra CNC hours at 15 ($\times 100$ TL) per hour. A temporary worker offers extra assembly hours at 15 ($\times 100$ TL) per hour. Assume the optimal basis does not change. Which offer(s) should the workshop accept? Explain using your answer to (b). 4 pts

Question 4 — Mathematical Foundations*15 points*

- (a) For $v = (1, -2, 2)$, compute $\|v\|_1$, $\|v\|_2$ and $\|v\|_\infty$. Sketch the unit ball $\{x \in \mathbb{R}^2 : \|x\|_1 \leq 1\}$. 3 pts

For parts (b)–(d), let

$$f(x, y) = x^2 - 2xy + 2y^2 + e^x$$

- (b) Compute the gradient $\nabla f(x, y)$ and the Hessian $\nabla^2 f(x, y)$. Evaluate both at $(0, 0)$. 4 pts

- (c) Prove that f is strictly convex on all of $\mathbb{R}^2$. State which convexity condition you are using. 4 pts

- (d) Write the second-order Taylor approximation $m(p) \approx f((0,0) + p)$ for $p = (p_1, p_2)$. 4 pts

Question 5 — Gradient Descent

15 points

Minimize $f(x) = x_1^2 + 5x_2^2$ starting from $x^{(0)} = (5, 1)$.

- (a) Write f in the form $\frac{1}{2}x^T Q x$. Give Q , its eigenvalues and the condition number $\kappa(Q)$.

2 pts

- (b) Take one gradient descent step with constant step size $\alpha = 0.1$. Give $x^{(1)}$ and $f(x^{(1)})$.

3 pts

- (c) For which constant step sizes α does gradient descent converge on this function? Explain what happens to each coordinate when $\alpha = 0.25$ is used for every iteration.

3 pts

- (d) Starting again from $x^{(0)}$, run **backtracking line search** along $d = -\nabla f(x^{(0)})$ with $\alpha_0 = 1$, $\beta = 0.5$ and Armijo constant $c = 0.1$. Write the Armijo condition, test every trial step and report the accepted α .

5 pts

- (e) With the step $\alpha = 2/(\lambda_{\min} + \lambda_{\max})$, by what factor does $\|x^{(k)} - x^*\|$ shrink per iteration (worst case)? Express the factor in terms of κ .

2 pts

Question 6 — Newton and Quasi-Newton Methods

10 points

(a) For the function f of Question 4, compute the Newton step Δx_{nt} at $(0, 0)$ and the Newton decrement $\lambda(0,0)$.

How much decrease in f does the quadratic model predict?

6 pts

(b) Why do quasi-Newton methods such as BFGS avoid computing $\nabla^2 f$? State the **secant condition** that the Hessian approximation B_{k+1} must satisfy and define every symbol in it.

4 pts

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Question 7 — Lagrange Multipliers and KKT

15 points

A sensor must be placed inside the allowed region $x + 2y \geq 5$, as close as possible to a base station at the origin:

$$\begin{array}{ll} \text{minimize} & x^2 + y^2 \\ \text{subject to} & x + 2y \geq 5 \end{array}$$

(a) Write the Lagrangian and all four KKT conditions for this problem.

4 pts

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(b) Solve the KKT system: find (x^*, y^*) , the multiplier μ^* and f^* . Show that the case $\mu = 0$ leads to no solution.

Explain why the KKT point is guaranteed to be the global minimum.

7 pts

(c) The right-hand side changes from 5 to 5.2. Using μ^* only (do not re-solve), estimate the new optimal value.

Then compare your estimate with the exact value.

4 pts

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— END OF EXAM · GOOD LUCK —